

Module 6

Git Basics for Course Work

Learning objectives

- Explain working tree, staging, and commits
- Use status, add, commit, log, diff
- Clone and push to GitHub
- Write a useful .gitignore for build artifacts

Prerequisites

- Module 5 (project structure)
- git installed

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Git model] --> B[init & clone]

B --> C[add & commit]

C --> D[remotes]

D --> E[gitignore]

E --> F[Module 7]

Course context

- Course work is submitted via Git
- Never commit build/ or large waveforms
- Small focused commits are easier to review

Core topics

Self-check

```
$ ./scripts/module6.sh --check
```

Terminal: module_check

```
#####  
Module 6: Environment self-check  
#####  
  
[OK] git available: git version 2.43.0  
[OK] module6 directory exists: /home/yongfu/proj/universal-verification-methodology/  
[OK] gitignore/sample.gitignore exists  
  
[OK] All required checks passed.
```

Basic workflow

```
git status  
git add file.sv  
git commit -m "Add counter testbench"
```


Git installation

```
$ git --version
```

```
Terminal: git_version  
git version 2.43.0
```

Exploring git help

```
$ git help -a 2>/dev/null | head -5
```

Terminal: git_help

See 'git help <command>' to read about a specific subcommand

Main Porcelain Commands

add	Add file contents to the index
am	Apply a series of patches from a mailbox

Practice

Exercise scaffold

```
./scripts/module6.sh --scaffold
```

Exercises

- Scaffold `git_demo` and make a commit
- Write `.gitignore` for `build/`
- `git log --oneline`

Summary & next steps

- Local Git workflow
- Remotes and GitHub
- Next: Module 7 — collaboration
- Next: Module 7: Advanced Git for Collaboration